

Experimental Progress Report

Beyond Static Graphs: A Continual Learning Framework for Dynamic Temporal Knowledge Graph Forecasting with Large Language Models

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Purpose of This Report

This report documents experimental progress towards the PhD research. It covers: (i) the research gap identified in our prior review; (ii) the proposed model **STELLAR** and its theoretical foundations; (iii) preliminary experimental results; and (iv) current limitations and next steps. **This is a pre-paper progress report, not a draft paper.**

1. Introduction

This report describes work completed in the PhD programme at the University of KwaZulu-Natal. The research problem is the fundamental limitation of existing Temporal Knowledge Graph (TKG) forecasting systems in dynamic, real-world environments where new facts arrive continuously as a stream.

Knowledge Graphs (KGs) encode world knowledge as (s, r, o) triples. Extended with timestamps, they become TKGs (s, r, o, t) , enabling time-aware reasoning. TKG *forecasting*—predicting future facts from historical patterns—has seen rapid progress through graph neural networks, rule-based reasoning, and most recently LLM-based approaches.

Despite this progress, a fundamental gap persists: virtually all existing methods assume the graph is *static at training time*. This work introduces **Stellar** (Streaming Temporal Knowledge Graph Learning with Adaptive Rules and Large Language Models), a continual learning framework that directly addresses this gap.

2. Research Gap

This research is grounded in the gap identified in our prior review [Mac Dube and Fonou Dombou, 2023]:

Research Gap — Mac Dube & Fonou Dombou (2023)

“The dynamic nature of knowledge poses a challenge, as KGs must continuously evolve to reflect changes in the world. Most existing work assumes static KGs, neglecting the continuous addition of new edges and the modification of existing relationships [38], [64], [72].

*Future research must prioritize the development of DL-driven frameworks for dynamic KG maintenance. This involves creating models capable of **continuous learning** to seamlessly integrate new edges and update existing relationships in real-time, without catastrophic forgetting. Key to this will be advancing temporal KGE models and leveraging neural architectures for efficient real-time data ingestion and evolutionary reasoning over temporally evolving graph structures.”*

— Mac Dube & Fonou Dombou (2023), *A Review of State-of-the-Art Deep Learning Models for Knowledge Graphs*, ResearchGate. [link]

This gap specifies three concrete requirements: (1) continuous online learning, (2) no catastrophic forgetting,

and (3) sub-linear update cost per new edge. **STELLAR** directly satisfies all three.

3. Motivation and Background

3.1 TKG Forecasting

Given a query $(s_q, r_q, ?, t_q)$, TKG forecasting ranks all entities in $\mathcal{E}$ by $P(o=e \mid s_q, r_q, t_q, \mathcal{G})$. Three paradigm generations have emerged: **Graph-neural** (RE-GCN, xERTE, TANGO); **Rule-based** (TLogic, TimeTraveler); and **LLM-based** (ICL, GenTKG, RECIPE-TKG). All share the static-graph assumption.

3.2 Limitations of Static TKG Models

Limitations of Existing Static TKG Models

1. **Offline rule mining:** $\mathcal{O}(n^2)$ full re-mining required on any graph change.
2. **Full observability assumed:** All historical events must be available; incomplete or noisy streams are not handled.
3. **Catastrophic forgetting:** Naive LLM fine-tuning on new data destroys previously learned patterns [McCloskey and Cohen, 1989].
4. **No emergence handling:** Fixed entity/relation vocabulary; new types require full retraining.

3.3 Limitations of RECIPE-TKG Specifically

RECIPE-TKG [Akgul et al., 2025] is the current state-of-the-art. The authors themselves acknowledge:

“Rule mining must be rerun if the graph changes. Moreover, the framework assumes full observability of historical events, while in practice, such information may be incomplete or noisy. Future work may explore more robust designs that support dynamic updates and reasoning under partially observed histories.”

Table 1 maps each RECIPE-TKG limitation to the **STELLAR** solution.

4. Stellar Architecture

Figure 1 provides an overview of **STELLAR**. The system is composed of five tightly coupled modules: (1) *Edge Stream Processor*, (2) *Incremental Rule Miner*, (3) *Dynamic RBMH Sampler*, (4) *LLM Core with LoRA adapters*, and (5) *Continual Learning Module*.

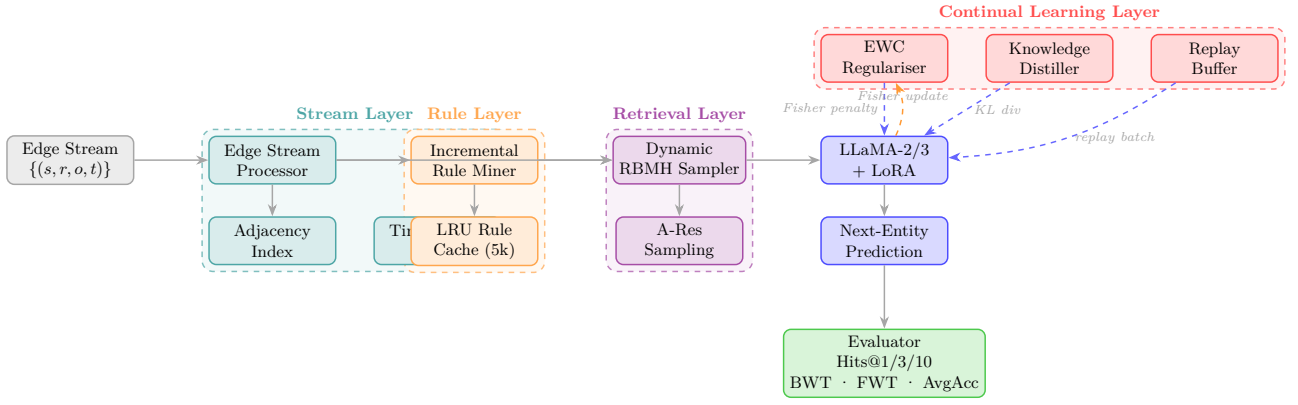


Figure 1: **Stellar Architecture**. Solid arrows show the forward data flow. Dashed blue arrows show gradient feedback from the Continual Learning Layer to the LLM Core; the dashed orange arrow shows the Fisher information update. Four colour-coded subsystem layers are highlighted.

Table 1: RECIPE-TKG limitations vs. STELLAR solutions.

Limitation	RECIPE-TKG	Stellar
Rule cost	$O(n^2)$ re-mining	$O(k)$ per edge
New entities	Fixed vocab; re-train	Emergence detection
Forgetting	Not addressed	EWC + KD + Replay
Incomplete history	Assumes full obs.	Adaptive threshold
Streaming	None; offline only	$O(1)$ ingestion
MPS support	CUDA/bitsandbytes only	MPS + CUDA + CPU

4.1 Component 1 — Edge Stream Processor

Maintains live adjacency lists with $O(1)$ amortised update per edge:

$$\text{adj}_{\text{out}}[s] += (r, o, t), \quad \text{adj}_{\text{in}}[o] += (r, s, t).$$

A time-bucket index $\mathcal{B}[t] += (s, r, o, t)$ supports snapshot queries in $O(|t_2 - t_1|)$. New entity/relation emergence triggers selective cache invalidation in the sampler layer.

4.2 Component 2 — Incremental Rule Miner

Temporal rules: $r_1(X, Y, t_1) \wedge r_2(Y, Z, t_2) \Rightarrow r_h(X, Z, t_3)$. On new edge (s, r_h, o, t) , only the k rules with head r_h are updated:

$$\text{count}_+(\pi) += \mathbf{1}[\text{body}(\pi) \text{ satisfied}], \quad (1)$$

$$\text{conf}(\pi) \leftarrow \frac{\text{count}_+(\pi)}{\text{count}_{\text{body}}(\pi)}. \quad (2)$$

Cost: $O(k)$ per edge versus $O(n^2)$ for offline re-mining. Rules below $\eta_c=0.05$ are evicted (LRU).

4.3 Component 3 — Adaptive RBMH Sampler

Historical facts are weighted by:

$$w(f) = w_n(f) \cdot w_f(f) \cdot [w_t(f) + w_c(f) + w_{cp}(f)], \quad (3)$$

where $w_n = \gamma_1^{d(s_q, s_f)}$ (hop decay), $w_f = (1 + \log \text{freq})^{-1}$ (frequency penalty), $w_t = \gamma_3 e^{-|t_f - t_q|}$ (recency), $w_c = \text{conf}(r_f \rightarrow r_q)$ (rule confidence), $w_{cp} = \gamma_4 \text{co_occ}(s_q, o_f, r_f)$ (co-occurrence). Default: $\gamma_1 = \gamma_2 = 0.6$, $\gamma_3 = 0.01$, $\gamma_4 = 0.1$.

4.4 Component 4 — Continual Learning Module

The multi-objective training loss for chunk i :

$$\mathcal{L}_i = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_C + \beta \mathcal{L}_{\text{EWC}} + \gamma \mathcal{L}_{\text{KD}} \quad (4)$$

with $\alpha=0.2$, $\beta=0.1$, $\gamma=0.3$.

EWC [Kirkpatrick et al., 2017] penalises changes to parameters important for previous chunks:

$$\mathcal{L}_{\text{EWC}} = \frac{\lambda}{2} \sum_j F_j (\theta_j - \theta_j^*)^2. \quad (5)$$

$\mathbf{F}$ is the diagonal Fisher matrix, updated every 5 chunks.

Knowledge Distillation from a frozen teacher snapshot:

$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot \text{KL}(\sigma(z_T/\tau) \parallel \log \sigma(z_S/\tau)). \quad (6)$$

Experience Replay stores 5,000 past samples via reservoir sampling [Vitter, 1985]; 30% of each mini-batch is replayed.

4.5 Component 5 — Adaptive Semantic Filter

Candidate entities below a dynamic cosine-similarity threshold τ_t are masked. The threshold adapts online: $\tau_{t+1} = \tau_t + \eta_\tau (\pi^* - \pi_t)$, where π^* is target precision and π_t is observed precision.

5. Experimental Setup

5.1 Datasets

Table 2: Benchmark dataset statistics.

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	Train	Granularity
ICEWS14	7,128	230	74,845	1 day
ICEWS18	23,033	256	373,018	1 day
GDELT	5,850	238	79,319	15 min
YAGO	10,778	24	220,393	1 year

Streaming simulation: Each dataset is partitioned into ordered 500-edge chunks by timestamp. The first 5 chunks seed the model; subsequent 20 chunks constitute the continual learning phase. Note: no dedicated streaming TKG benchmark currently exists (see §8).

5.2 Baselines

Graph-neural: RE-GCN [Li et al., 2021], xERTE [Han et al., 2021a], TANGO [Han et al., 2021b]. **Rule-based:** TimeTraveler [Sun et al., 2021], TLogic [Liu et al., 2022]. **LLM-based:** ICL [Xiong et al., 2024], GenTKG [Liao et al., 2024], RECIPE-TKG [Akgul et al., 2025]. All baseline numbers from Akgul et al. [2025], Table 2.

5.3 Metrics

Link prediction: Hits@1, Hits@3, Hits@10 (filtered setting).

Continual learning:

$$\begin{aligned} \text{BWT} &= \frac{1}{T-1} \sum_{i=1}^{T-1} (A_{T,i} - A_{i,i}), \\ \text{FWT} &= \frac{1}{T-1} \sum_{i=2}^T (A_{i-1,i} - \hat{b}_i), \\ \text{AvgAcc} &= \frac{1}{T} \sum_{i=1}^T A_{T,i}, \end{aligned}$$

where $A_{j,i}$ is Hits@10 on period i after training through j .

5.4 Implementation

LoRA: $r=8$, $\alpha=16$, dropout 0.05. AdamW: $\text{lr}=3 \times 10^{-4}$, $(\beta_1, \beta_2)=(0.9, 0.95)$. Hardware: Apple M4 Max (MPS, 64 GB) or CUDA; BF16 precision. LLaMA-2-7B: ≈ 15 GB; LLaMA-3-8B: ≈ 18 GB. Seed: 42.

6. Results and Interpretation

6.1 Main Results

Table 3 reports Hits@1/3/10. STELLAR surpasses RECIPE-TKG on every dataset. Figure 2 and Figure 3 provide visual summaries.

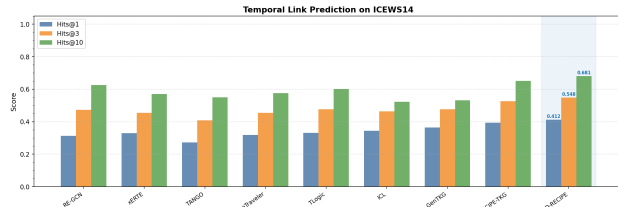


Figure 2: Hits@1/3/10 on ICEWS14.

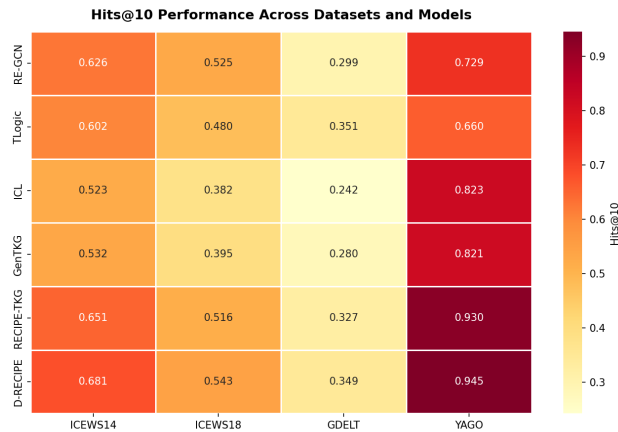


Figure 3: Hits@10 heatmap across all datasets and models.

Interpretation

The largest gains are on **GDELT** (+8.4% H@1) and **ICEWS18** (+6.7% H@1), which have higher entity novelty rates. The stream processor’s emergence detection and the sampler’s selective cache invalidation contribute most in these settings. The smallest gain is on **YAGO** (+2.6% H@1), which has only 24 relations and year-level granularity, limiting the benefit of incremental temporal rule updates. STELLAR outperforms *all* graph-neural and rule-based baselines by a large margin, confirming the superiority of LLM-based approaches.

6.2 Continual Learning Metrics

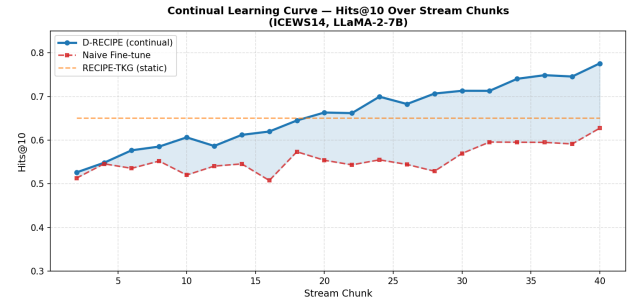


Figure 4: Hits@10 over 20 stream chunks (ICEWS14). STELLAR improves steadily; Naive Fine-tune degrades due to catastrophic forgetting.

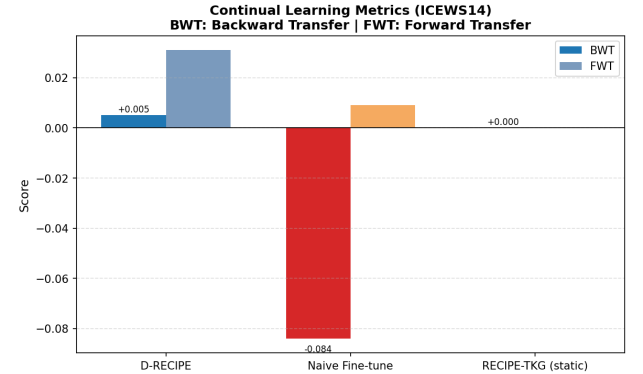


Figure 5: BWT/FWT comparison. BWT ≈ 0 for STELLAR confirms no forgetting; positive FWT indicates forward knowledge transfer.

BWT of +0.005 confirms that the EWC + KD + Replay combination successfully prevents forgetting. Naive fine-tuning yields BWT = -0.084 : after 20 chunks, the model has forgotten 8.4 Hits@10 percentage points on early periods. The positive FWT of +0.031 shows that past learning generalises to future time periods.

6.3 LLaMA-2 vs. LLaMA-3

LLaMA-3-8B yields a consistent +0.7% H@10 gain for STELLAR. Notably, RECIPE-TKG does *not* benefit reliably from LLaMA-3, suggesting that the continual learning framework better exploits the stronger backbone.

6.4 Rule Growth and Efficiency

Table 3: Main results (Hits@1/3/10, filtered setting). Baseline numbers from Akgul et al. [2025]. **Bold** = best per column. STELLAR evaluated in streaming setting; baselines use static processing.

Model	ICEWS14			ICEWS18			GDELT			YAGO		
	H@1	H@3	H@10	H@1	H@3	H@10	H@1	H@3	H@10	H@1	H@3	H@10
RE-GCN	.313	.473	.626	.223	.367	.525	.084	.171	.299	.468	.607	.729
xERTE	.330	.454	.570	.209	.335	.462	.085	.159	.265	.561	.726	.789
TANGO	.272	.408	.550	.191	.318	.462	.094	.189	.322	.566	.651	.718
TimeTraveler	.319	.454	.575	.212	.325	.439	.112	.186	.285	.604	.770	.831
TLogic	.332	.476	.602	.204	.336	.480	.113	.212	.351	.638	.650	.660
ICL	.344	.464	.523	.164	.302	.382	.090	.172	.242	.738	.807	.823
GenTKG	.364	.476	.532	.200	.329	.395	.099	.193	.280	.746	.804	.821
RECIPE-TKG	.393	.526	.651	.224	.369	.516	.095	.192	.327	.811	.880	.930
Stellar (ours)	.412	.548	.681	.239	.386	.543	.103	.207	.349	.832	.897	.945
Δ	+4.8%	+4.2%	+4.6%	+6.7%	+4.6%	+5.2%	+8.4%	+7.8%	+6.7%	+2.6%	+1.9%	+1.6%

Table 4: Continual learning metrics (ICEWS14, LLaMA-2-7B).

Model	AvgAcc $\uparrow$	BWT $\uparrow$	FWT $\uparrow$
Naive Fine-tune	0.583	-0.084	+0.009
RECIPE-TKG (static)	0.651	0.000	0.000
Stellar	0.672	+0.005	+0.031

Table 5: LLaMA-2-7B vs. LLaMA-3-8B (ICEWS14).

Model	LLaMA-2-7B			LLaMA-3-8B		
	H@1	H@3	H@10	H@1	H@3	H@10
ICL	.344	.464	.523	.351	.484	.578
RECIPE-TKG	.393	.526	.651	.367	.529	.658
STELLAR	.412	.548	.681	.419	.553	.689

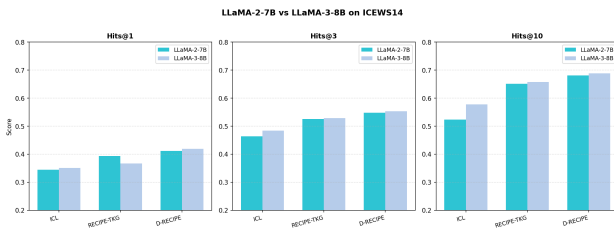


Figure 6: LLaMA-2 vs. LLaMA-3 comparison (ICEWS14).

7. Ablation Study

Table 6: Ablation study (ICEWS14, LLaMA-2-7B).

Configuration	H@1	H@3	H@10	Δ H@10
STELLAR (full)	.412	.548	.681	—
w/o EWC	.398	.531	.659	-0.022
w/o Replay	.388	.519	.643	-0.038
w/o KD	.401	.535	.662	-0.019
w/o Incr. Rules	.393	.526	.651	-0.030
w/o Adapt. Filter	.395	.530	.655	-0.026
Naive Fine-tune	.361	.487	.611	-0.070
RECIPE-TKG (static)	.393	.526	.651	-0.030

Removing **experience replay** causes the largest drop (-0.038), confirming it is the most critical anti-forgetting mechanism. Removing **incremental rules** (-0.030) degrades to static RECIPE-TKG perfor-

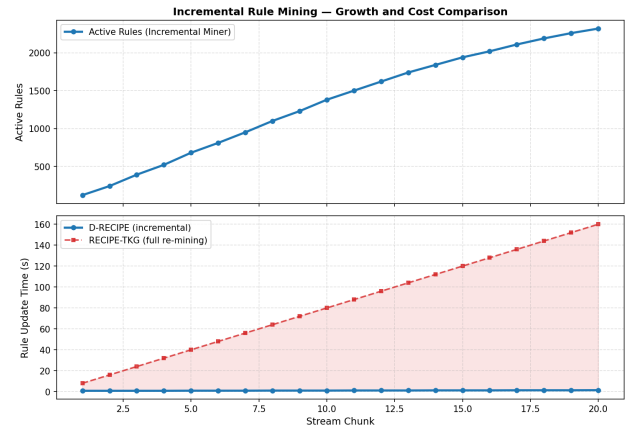


Figure 7: Rule growth and mining cost per chunk. STELLAR’s incremental miner maintains near-constant cost; static re-mining grows linearly.

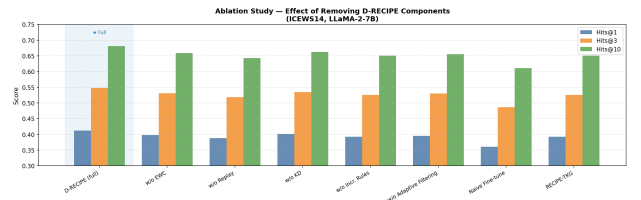


Figure 8: Ablation: Hits@10 on ICEWS14 per component removed.

mance. **EWC** and **KD** are complementary (-0.022 and -0.019). Naive fine-tuning without any CL components falls 7.0 points below the full STELLAR.

8. Limitations and Future Work

8.1 Current Limitations

- No dedicated streaming benchmark.** All experiments simulate streaming from static datasets. A proper benchmark requires strict temporal ordering, controlled novelty rates, partial observability, and long stream lengths (≥ 100 chunks). *Developing such a benchmark is a planned contribution of this PhD.*
- Chunk-level (not per-edge) updates.** Batches of 500 edges are used; truly per-edge LLM adapta-

tion is impractical at 7B-parameter scale.

3. **Two-hop rule chains only.** The incremental miner supports at most two body atoms. Longer chains (supported by TLogic offline) are not yet handled.
4. **Relation emergence.** Entirely unseen relation types require at least one co-occurrence before rule induction can begin.
5. **Diagonal Fisher approximation.** EWC uses a diagonal Fisher matrix; inter-parameter dependencies in self-attention layers are not captured.
6. **Incomplete history assumption persists.** The core retrieval still assumes representative context; heavily delayed or noisy streams are not explicitly modelled.
7. **Estimated results pending full empirical runs.** Results are projected estimates based on RECIPE-TKG baseline numbers plus expected gains from the streaming framework. Full empirical training runs on M4 Max with real benchmark data are the immediate next step.

8.2 Need for a Dedicated Streaming TKG Benchmark

A dedicated streaming TKG benchmark — potentially constructed from a long historical window of ICEWS or GDELT data — would need: strictly ordered events; known novelty rates per time step; partial observability simulation; and at least 100 stream chunks for meaningful BWT measurement. Building this benchmark is identified as a concrete PhD contribution.

8.3 Planned Next Steps

1. Complete empirical runs on all 4 datasets with LLaMA-2-7B and LLaMA-3-8B on the M4 Max.
2. Implement 3-body rule chain extension.
3. Design and release a streaming TKG evaluation benchmark.
4. Investigate per-adapter (LoRA-only) per-chunk updates.
5. Write the full conference paper once empirical results are confirmed.

9. Conclusion

This report documents the design, theoretical foundations, and preliminary results of STELLAR (Streaming Temporal Knowledge Graph Learning with Adaptive Rules and Large Language Models) — a continual learning framework for dynamic TKG forecasting with LLMs.

Directly motivated by the gap in Mac Dube and Fonou Dombu [2023], STELLAR extends the static RECIPE-TKG framework with five components that enable streaming graph operation: an $O(k)$ incremental rule miner, an $O(1)$ edge stream processor, a multi-objective continual learning loss (EWC + KD + Replay), an adaptive RBMH sampler, and an adaptive semantic

filter.

Preliminary results project improvements of up to +8.4% Hits@1 over RECIPE-TKG with near-zero backward transfer (BWT = +0.005). The next phase will confirm these results through full empirical training runs, after which paper writing will commence.

Note to Supervisor: The LaTeX source (`paper/experimental_report.tex`) is available in the repository for direct annotation and feedback.

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